Experiment-9 (double linked list)

Q1) Write a C program to perform the operations on a Double linked list:

i) Insertion at beginning, ii) Deletion of 1st node iii) display all nodes

```
#include<stdio.h>
#include<stdlib.h>
struct node{
   int data;
   struct node *prev;
   struct node *next;
};
struct node *head=NULL;
void insert_begin(){
   struct node *temp;
   int x;
   temp=(struct node*)malloc(sizeof(struct node));
   printf("Enter value: ");
   scanf("%d",&x);
   temp->data=x;
   temp->prev=NULL;
   temp->next=head;
   if(head!=NULL)
      head->prev=temp;
   head=temp;
}
void delete_first(){
   struct node *temp;
   if(head==NULL){
      printf("List empty\n");
      return;
```

```
    }
    temp=head;
    head=head->next;
    if(head!=NULL)
        head->prev=NULL;
    free(temp);
}
void display(){
    struct node *temp=head;
    if(head==NULL){
        printf("List empty\n");
        return;
    }
    while(temp!=NULL){
        printf("%d ",temp->data);
        temp=temp->next;
    }  printf("\n");
}
int main(){
    int ch;
    while(1){
        printf("\n1.Insert Beginning\n2.Delete First\n3.Display\n4.Exit\n");
        scanf("%d",&ch);
        switch(ch){
            case 1: insert_begin(); break;
            case 2: delete_first(); break;
            case 3: display(); break;
            case 4: exit(0);
        }
    }   }
```

Q2) Write a C program to perform the operations on a double linked list:

i) insertion at end, i) deletion of last node iii) display all the nodes

```
#include<stdio.h>
#include<stdlib.h>
struct node{
   int data;
   struct node *prev;
   struct node *next;
};
struct node *head=NULL;
void insert_end(){
   struct node *temp,*ptr;
   int x;
   temp=(struct node*)malloc(sizeof(struct node));
   printf("Enter value: ");
   scanf("%d",&x);
   temp->data=x;
   temp->next=NULL;
   if(head==NULL){
      temp->prev=NULL;
      head=temp;
      return;
   }
   ptr=head;
   while(ptr->next!=NULL)
      ptr=ptr->next;
   ptr->next=temp;
   temp->prev=ptr;
}
```

```
void delete_last(){
    struct node *temp=head;
    if(head==NULL){
        printf("List empty\n");
        return;
    }
    if(head->next==NULL){
        free(head);
        head=NULL;
        return;
    }
    while(temp->next!=NULL)
        temp=temp->next;

    temp->prev->next=NULL;
    free(temp);
}
void display(){
    struct node *temp=head;
    if(head==NULL){
        printf("List empty\n");
        return;
    }
    while(temp!=NULL){
        printf("%d ",temp->data);
        temp=temp->next;
    }
    printf("\n");
}
int main(){
```

```
    int ch;
    while(1){
        printf("\n1.Insert End\n2.Delete Last\n3.Display\n4.Exit\n");
        scanf("%d",&ch);
        switch(ch){
            case 1: insert_end(); break;
            case 2: delete_last(); break;
            case 3: display(); break;
            case 4: exit(0);
        }
    }
}
```